Base de Données de Section Efficaces d'Ionisation (BDSEI): A Database Of Low Energy Total Ionisation Cross-Section For Astrochemical Purpose

Lilian Heydel[1]

[1]Centre de Recherche sur les Ions, les Matériaux et la Photonique Université Caen Normandie, GANIL, ENSICAEN, CNRS, CEA, Normandie Univ, CIMAP UMR6252, F-14000 Caen, France. **lilian.heydel@ganil.fr**

Total Ionisation Cross Section (TICS) reprents a certainly important factor for a large number of physical domains.

1 Introduction

Electron-driven ionisation plays a big role in planetary and interplanetary medium. The result of it the production of reactive radicals. One of the main parameter to quantify this phenomena is the cross-section of ionisation of the molecule, either partial (Partial Ionisation Cross Section, **PICS**) or total (Total Ionisation Cross Section, **TICS**). While it can be mesured experimentally (Rapp and Englander-Golden, 1965; Jiao, DeJoseph, and Garscadden, 2007, and reference therein), this is not necesseraly the case for every molecules. In fact, some molecules are either absent or under covered litterature-wise. Those include molecule of interest for planetary and exoplanetary science, like phosphine PH_3, found in the atmosphere of gas giants and in the interstellar medium (Ridgway, Wallace, and Smith, 1976; Larson et al., 1980; Burgasser et al., 2025), propaenitril C_3H_5N, found in Titan atmosphere (cite), or even brome bearing molecules such as ..., ..., ... The same can be said for isomers, as not all isomer of a define specie have the TICS calculated for all its isomer. It is, however, possible to compute this parameter. Total Ionisation Cross-Section (TICS) database established using computing method were already done, such as ALeCS (Gaches et al., 2024), NIFS (citer) or NIST (citer). They, however, lack the presence of some of those *exotic* specie, or are either complicated to acces for general public.
In this work, I present a database of TICS for 274 neutral and stable molecules made with open-source tools. This database is also available on a website hosted on GitHub and is publicly available[1]. All the content of the website can be downloaded freely.

2 Methodology

The methodology employed in this work is based around the geometry of each molecules.

2.1 Molecular geometry pipeline

The most extensive database for 3D molecular structure is *PubChem* (cite). The code make a request at PubChem server with the molecule name and fetch for the 3D structure if available, are the 2D in last resort. The geometry is then downloaded in the `sdf` format and converted into the `xyz` format. The 3D geometry catalogue of Pubchem was calculated using repulsion method and should be already optimal; geometry optimisation is pratically useless in that case (see appendix **??**). One problem emerge regarding molecules with no 3D geometry available; those include molecules of astrochemical interest, such as silane SiH_4 (found in the atmosphere of explanets; <**empty citation**>) or sulfane H_2S (found on the cloud deck of ice giants; <**empty citation**>). In that case, starting calculation with 2D structure was proven to be false. Thankfully, it is possible to obtain 3D geometry from another way. Using the `strainedSMILES2xyz` python package (Demuth, Schnizer, and Svatunek, 2026), we can retrieve a coherent 3D structure from the SMILES structure. This generated geometry can be then atomically optimized using the `geometric` package (Wang and Song, 2016) for several cycles, in order to get coherent geometry.

2.2 The BEB method for TICS calculation

The *Binary Encounter Bethe* (BEB) method was proposed by Kim and Rudd (1994) in order to provide a simplier model than the *Binary Encounter Dipole* (BED) for ionisation cross-sections calculation. The BEB method estimates the cross-section of total ionisation using the electron energy of the MO and the kinetic energy of the electron. In application, the BEB method can expressed as, for a distinct MO i:

$$\sigma_i(T) = \frac{S}{t+u+1}\left[\frac{\ln t}{2}\left(1-\frac{1}{t^2}\right)+1-\frac{1}{t}-\frac{\ln t}{t+1}\right] \tag{1}$$

with $t = \frac{T}{B}$, $u = \frac{U}{B}$, where T the ionisation particle energy and B and U are the aforementioned MO electrons and kinetic energy, and S is expressed as:

$$S = 4\pi a_0^2 N_i \left(\frac{R}{B}\right)^2 \tag{2}$$

with a_0 the Bohr radius (0.5292 $\mathring{A}$), R the Rydberg constant (13.6 eV) and N_i the number of electrons in the MO. For neutral stable species, N is always equal to 2. This, however, gives the cross section for only one MO. To get the total ionisation cross section, we need to sum all the partial ionisation cross-section of each MO,

[1]bdsei.com mais en vrai j'ai tjrs pas uploadé donc faut attendre pour avoir la vraie url

Table 1: List of molecules present in the database. Superscript number indicates the number of isomers.

Number of atoms													
1	2	3	4	5	6	7	8	9	10	11	12	13	≥ 14
Ar	C_2	C_2O	C_2H_2	C_3HN^2	C_4H_2	C_5HN	C_6H_2	$C_5H_4^2$	C_8H_2	C_5H_5N	$C_6H_6^3$	C_7H_5N	$C_{25}H_{11}N$
He	CO	CHN^2	$C_2N_2^2$	C_2H_2O	C_4N_2	C_3H_3N	C_6N_2	C_4H_4S	$C_4H_4N_2^2$	C_5H_6	$C_4H_6O_2^2$	$C_5H_8^5$	$C_{17}H_9N^3$
S	CS	CHP	CH_2O	C_2H_2S	$C_3H_2O^3$	$C_3H_4^3$	$C_4H_3N^3$	$C_3H_4N_2^2$	$C_4H_5N^2$	C_4H_6O	$C_4H_7N^2$	$C_4H_8O^2$	$C_{16}H_{10}$
	CSi	CKN	CH_2S	C_2HNO	C_2Cl_4	$C_2H_3NO^2$	C_4H_4	C_3H_5N	$C_4H_6^4$	$C_3H_6O_2^2$	$C_4H_8^5$	C_4H_8S	$C_{14}H_{10}$
	AlCl	CNNa	$CHNO^3$	CCl_4	$C_2H_2O_2$	$C_2H_4O^3$	$C_3H_3NO^3$	$C_3H_6^2$	C_3H_5NO	C_3H_7N	$C_3H_6O_3$	$C_3H_8O_2^2$	$C_{11}H_{12}N_2O_2$
	AlF	CO_2	$CHNS^3$	CF_4	C_2H_3Cl	$C_2H_4S^2$	C_3H_3NS	$C_2H_5NO^2$	$C_3H_6O^4$	C_3H_8	C_3H_7NO	$C_3H_9N^3$	$C_{11}H_7N^2$
	Br_2	COS	AsH_3	CH_2Cl_2	$C_2H_3N^3$	CH_4Si_2	$C_3H_4O^2$	$C_2H_6O^2$	$C_2H_5NO_2$	C_2H_7NO	$C_3H_8O^3$	C_3H_9P	$C_{10}H_8$
	BrH	CS_2	H_2O_2	CH_2F_2	C_2H_4	CH_5N	$C_3H_4N_2$	$C_2H_6S^2$	$C_2H_6O_2^2$				C_9H_8
	CaO	Cl_2O	H_3N	$CH_2N_2^2$	C_2HCl_3	AlH_3O_3	$C_2H_4O_2^4$		C_2H_6OS				$C_8H_{10}^3$
	Cl_2	ClNO	H_3P	CH_2O_2	CH_2O_3	F_6S	$C_2H_5N^2$		$C_2H_7N^2$				C_8H_8
	ClH	FNO	H_3Sb	CH_2S_2	CH_3NO	H_5NS	C_2H_6						$C_7H_8^2$
	ClK	H_2O	HNO_2	CH_3Br	CH_3NS	N_2O_5	CH_3ClO_2S						C_6H_{10}
	ClNa	H_2S	O_3S	CH_3Cl	CH_3NSi		CH_4N_2O						$C_6H_{12}^9$
	F_2	H_2Se		CH_3F	CH_4O		H_6Si_2						$C_6H_{14}^4$
	FH	HNO		CH_3I	CH_4S								$C_5H_{10}^3$
	FeO	HNaO		CH_3P	H_4N_2								C_5H_{12}
	H_2^2	N_2O		CH_4									C_5H_8O
	HI	O_2S		$CHCl_3$									$C_4H_{10}^2$
	HP	O_2Ti		CHF_3									$C_4H_{10}O^4$
	MgS	O_3		GeH_4									$C_4H_{10}S$
	N_2			H_3NO									$C_4H_8O_2^3$
	O_2			H_4Si									$C_3H_8O_3$
	OS												
	OSi												
	OTi												

which gives us:

$$\sigma_{TICS}(T) = \sum_{i=1}^{n} \sigma_i(T) \tag{3}$$

Concerning T, this parameter is kept at relatively low energy. We considered 10^3 values in a logscale array with $T \in [10^1; 10^5]$. It is possible to upgrade this method by inserting several coefficient and functions, as seen in Gaches et al. (2024). This is not the case here as we kept the method as close as equation 1.

2.3 Quantum chemistry parameters calculations

The cross-section calculations, discussed in subsection **??**, requires molecular orbits (MO) energies, either electrons (B) or kinetic (U) energies. While NIST database (cite) propose this parameter for several molecule, this is limited in terms of selection. One way is to calculate these parameter. To do that, a model of quantum chemistry have been employed. This model is `pyscf` (Sun et al., 2017), an open-source python library that have been used in several published papers in the past (cite). Using the downloaded geometries, we can input them into the model and obtain the required paramaters. This model is based on the Self Consistent Field (SCF) method of quantum chemistry and resolves the Hatree-Focke (HF) solution. The basis-set family used in this work is the `def2-TZVPPD` basis-set – valence triple-zeta with sets of polarization and diffuse functions, proposed by Weigend and Ahlrichs (2005).This type of basis, in particular the `def2-TZVPP` basis, is proven to be reliable for HF modeling for a majority of the periodic table, from H up to Rn (Zheng, Xu, and Truhlar, 2011; Hellweg and Rappoport, 2015). This enables us to cover a wide range of molecular species using a single basis-set while other basis like `cc-` (Dunning, 1989), requires a specialized basis for that kind of work. Several BEB-related papers use either cross-correlated `cc-` base (Kumar, 2014; Gaches et al., 2024), proposed by Dunning (1989) or gaussian `6-311G` basis (Ali et al., 1997), proposed by Krishnan et al. (1980). At our knowledge, this is the first use of a `def2-` basis for TICS calculations.

The BEB method, by itself, has limitations for heavy elements, with quantum number $n > 2$ or $Z > 10$ (Huo and Kim, 2000; Scott and Irikura, 2005; Ali and Kim, 2008; Graves, Cooper, and Tennyson, 2021). The discrepancy arises because the BEB method uses, as shown in Eq. 1, the kinetic energy of each molecular orbital as a variable. This is problematic for heavier elements, as inner electronic layers of this type have very high kinetic energies. However, ionisation should occur when the electron is far from the centre of the shell, i.e. when it is slower and thus has a lower kinetic energy. An overestimation of the kinetic energy leads to an underestimation of σ_{TICS}. The solution proposed by Huo and Kim (2000), and employed in several BEB-related studies on heavy elements (Scott and Irikura, 2005; Dolg and Cao, 2012; Graves, Cooper, and Tennyson, 2021), is to use the effective core potential (ECP) method. An ECP limits the interaction of valence electrons with the core shell by replacing the latter with an effective potential, while the valence electrons are described by pseudo-valence orbitals that do not penetrate the inner region of the nucleus. In a nut*shell*, the ECP removes the high-kinetic-energy core orbitals from the BEB sum by using the closest rare gases, which otherwise artificially suppress the valence contribution. The ECP method can be used in `pyscf` and, in order to include as many molecules as possible in the database, was implemented in this work. This, however, requires a different basis than `def2-TZVPPD`, because it's ECP basis only start at Rb ($Z \geq 37$). While ECP-orbital basis have been developped in the mid 1980's for both $3p$ (Fernandez Pacios and Christiansen, 1985) and $4p$ (Hurley et al., 1986) -the `crenbl` basis combo, it is rather rare to have modern ECP basis for lighter elements. Thank-

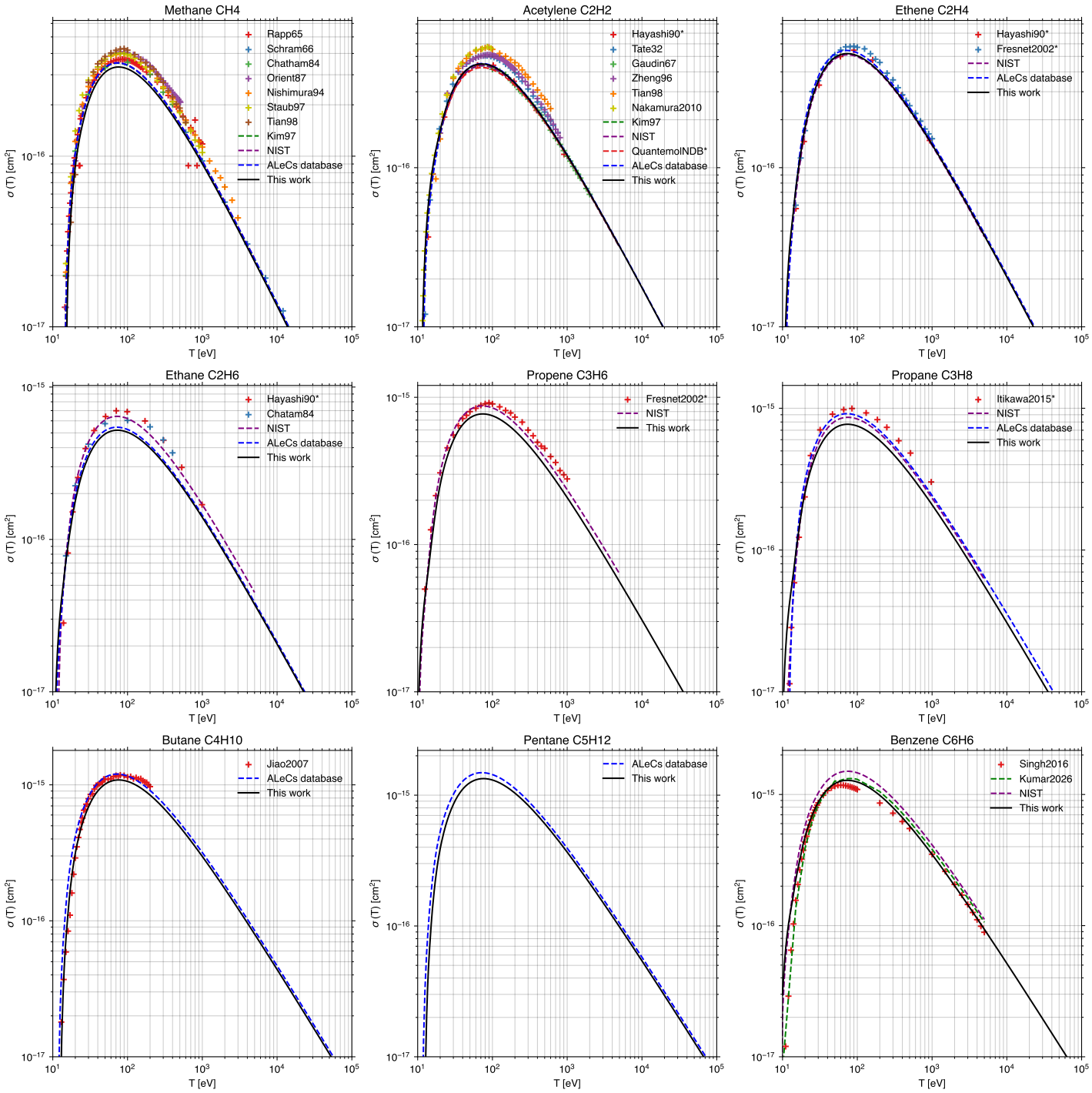


Figure 1: Calculted TICS for several hydrocarbons. In those plots, σ_{TICS} as been cropped at 10^{-8}. Scatter plot mean experimental data and line plot mean theoritical data. Those data come from Rapp and Englander-Golden (1965), Chatham et al. (1984), Ali et al. (1997), and Jiao, DeJoseph, and Garscadden (2007). The references marked with an * where obtained through lxcat.com (<**empty citation**>) database. Those references include **bibid**.

fully, the open-source `pseudopotentiallibrary.org`[2] provides Dunning derived basis combo for both $3p$ and $4p$ elements. For Al, Si, P, S, Cl and Ar bearing molecules, we used the `aug-cc-pVTZ` orbital basis along with its associated ECP basis proposed by (Bennett et al., 2018), whereas for Ge, As, Se and Br bearing molecules we used the same orbital basis and the ECP basis proposed by (Wang et al., 2019). The counter-effect of this approach is that higher energies cross sections tends to be underestimated. But it is worth for having a better peak result. Figure **??** in appendix **??** presents a comparison

[2] https://pseudopotentiallibrary.org/

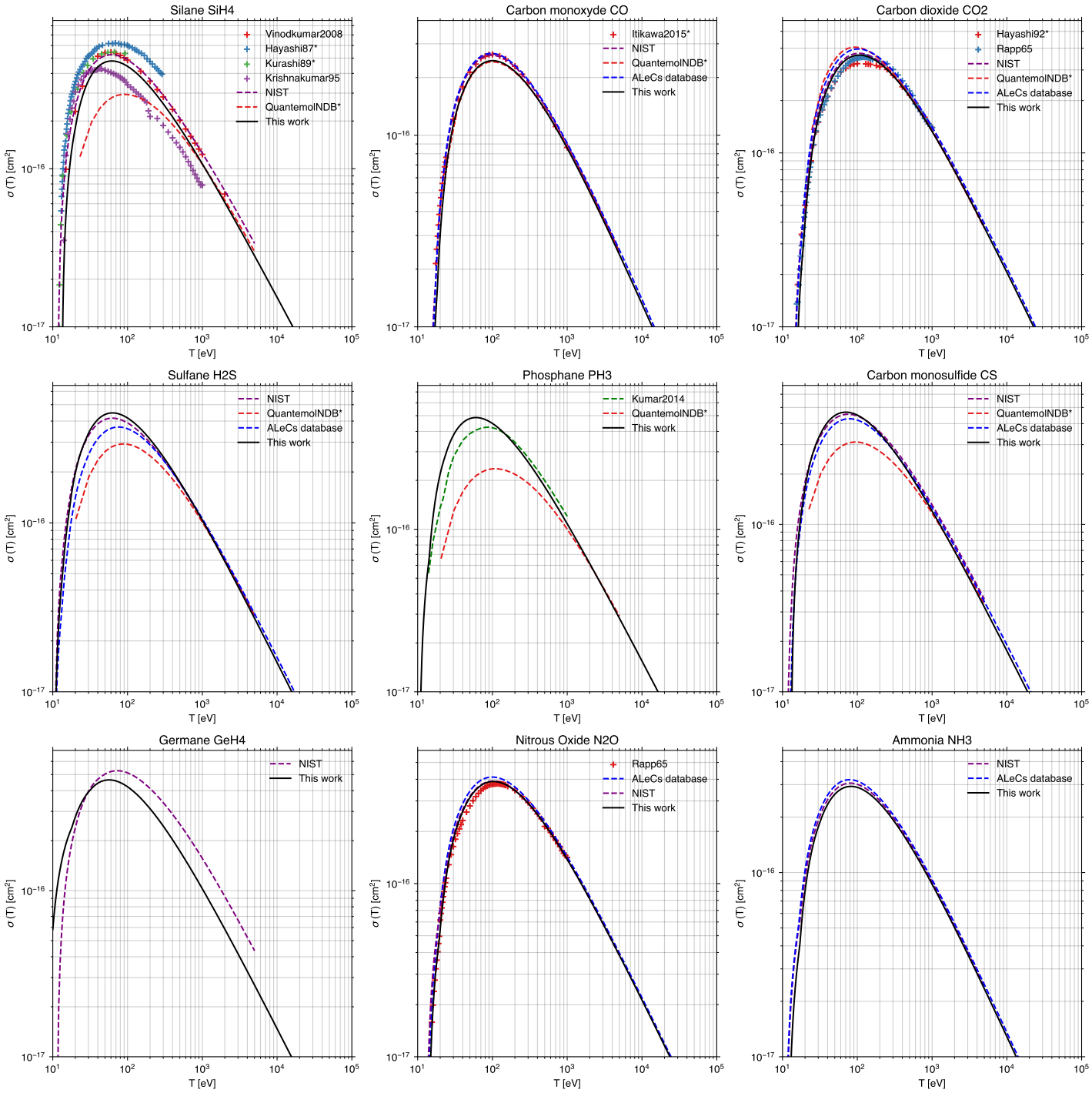


Figure 2: Calculted TICS for several organic and non-organic molecules. In those plots, σ_{TICS} as been cropped at 10^{-8}. Scatter plot mean experimental data and line plot mean theoritical data. Those data come from Rapp and Englander-Golden (1965), Chatham et al. (1984), Ali et al. (1997), Jiao, DeJoseph, and Garscadden (2007), and Kumar (2014). The references marked with an * where obtained through lxcat.com (<**empty citation**>) database. Those references include **bibid**.

3 Results

3.1 Overview

274 ionisation cross section of unique molecules have been calculated, and up to 10 isomer for some formula[3]. The complete list is visible on table 1.

[3]Like C_6H_{12} where 10 isomers are available.

3.2 ECP impact on TICS calculation

As seen in Huo and Kim (2000), enabling ECP for BEB calculations indeed increase the TICS for molecule containing $Z > 10$ elements.

Table 2: sqdsd

	$\overline{\Delta\sigma}$ (%)	
Molecule	cc-ECP	CRENBL
H3P	13.996	14.568
H4Si	6.505	7.311
H6Si2	9.242	9.283
Geh4	27.548	32.212
H2Se	27.384	31.159
H2S	18.474	16.958
Ch3Br	20.059	22.418
Brh	24.582	27.666
Br2	26.276	26.496
Cs	11.996	0.948
Cl2O	18.123	12.592
Osi	5.703	6.007
Csi	7.079	7.372
Ch4Si2	5.093	7.995
Ch3Nsi	2.749	3.753
C4H10S	9.545	9.718

4 Discussion and comparison with litterature

4.1 Overview

Des soucis avec les hydrocarbures saturés ? Le propene par ex

First and foremost, experimental B values tend to be smaller than theoritical/computed B values (Hwang, Kim, and Rudd, 1996; Ali et al., 1997). As such, the cross section obtained through experimental B values tend to be higher,≈10-15% at the cross section peak. This was observed several time in the plots of figures 1-3, where theoritical curves were generally smaller than those from experimental papers. This is not systemic.

4.2 Limits

Whereas this paper serves as a database presentation, some results do harbour scientific interest, as some molecule haven't got a dedicated TICS paper yet. blalbalbalb

As shown in several studies, isomery play a minor role regarding σ_{TICS}. dshfdv

Ici évoque le SF6, qui a un petit intérêt astro mais que j'arrive pas a calcluer la géométrie

5 Conclusion

Pouvoir utiliser un spectre complet de section efficace d'ionisation comme ceux que j'ai calculé pour calculer la section effiace d'ionisation partielle ? Pouvoir brancher cela a des autres modeles (comme celui de Gautier 2020) ? Faire du calcul de flux d'ionisation, jpeux essayer de lier ca a triton

Multiple outlook could be imaginated:

- Adapt the pipeline for radicals (*i.e* molecules with spin ≠ 0) and ions; This would require to do calculation using the UHF method of SCF;

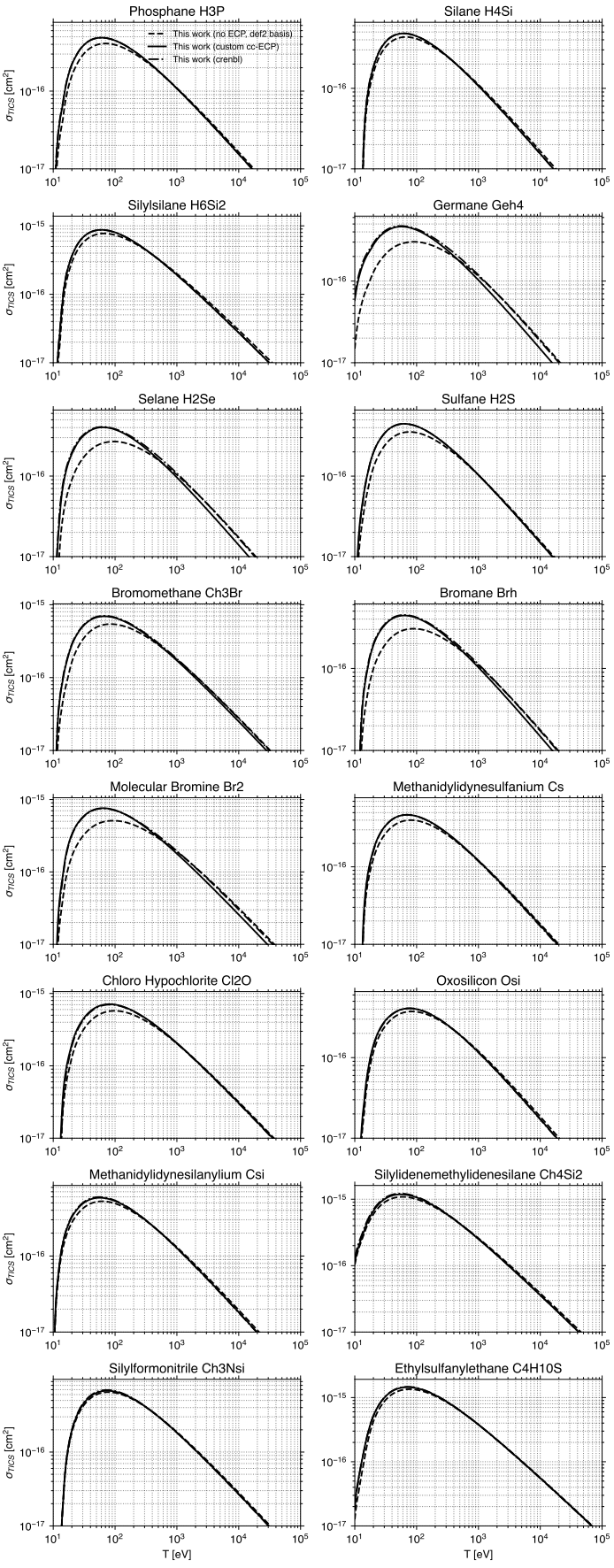


Figure 3: Comparison of TICS results for several molecule with and without ECP enabled

- Refine and optimize the geometry of molecules -if possible. This could be useful for molecules taken from a 2D structure.
- Adapt and create a custom `def2-` ECP basis for $3d$ and $4p$ elements in order to homogenize the database as much as possible protocol-wise;

Acknowledgement

References

Rapp, Donald and Paula Englander-Golden (Sept. 1965). "Total Cross Sections for Ionization and Attachment in Gases by Electron Impact. I. Positive Ionization". In: *The Journal of Chemical Physics* 43.5, pp. 1464–1479. ISSN: 0021-9606. DOI: 10.1063/1.1696957. (Visited on 10/27/2026).

Ridgway, S. T., L. Wallace, and G. R. Smith (Aug. 1976). "The 800 - 1200 Inverse Centimeter Absorption Spectrum of Jupiter." In: *The Astrophysical Journal* 207, pp. 1002–1006. ISSN: 0004-637X. DOI: 10.1086/154570. (Visited on 10/28/2026).

Krishnan, R. et al. (Jan. 1980). "Self-Consistent Molecular Orbital Methods. XX. A Basis Set for Correlated Wave Functions". In: *The Journal of Chemical Physics* 72.1, pp. 650–654. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.438955. (Visited on 10/28/2026).

Larson, H. P. et al. (Aug. 1980). "The Middle-Infrared Spectrum of Saturn - Evidence for Phosphine and Upper Limits to Other Trace Atmospheric Constituents". In: *The Astrophysical Journal* 240, pp. 327–337. ISSN: 0004-637X. DOI: 10.1086/158236. (Visited on 10/28/2026).

Chatham, H. et al. (Aug. 1984). "Total and Partial Electron Collisional Ionization Cross Sections for CH4, C2H6, SiH4, and Si2H6". In: *The Journal of Chemical Physics* 81.4, pp. 1770–1777. ISSN: 0021-9606. DOI: 10.1063/1.447848. (Visited on 10/22/2026).

Fernandez Pacios, Luis and P. A. Christiansen (Mar. 1985). "Ab Initio Relativistic Effective Potentials with Spin-orbit Operators. I. Li through Ar". In: *The Journal of Chemical Physics* 82.6, pp. 2664–2671. ISSN: 0021-9606. DOI: 10.1063/1.448263. (Visited on 10/28/2026).

Hurley, M. M. et al. (June 1986). "Ab Initio Relativistic Effective Potentials with Spin-orbit Operators. II. K through Kr". In: *The Journal of Chemical Physics* 84.12, pp. 6840–6853. ISSN: 0021-9606. DOI: 10.1063/1.450689. (Visited on 10/29/2026).

Dunning, Thom H. (Jan. 1989). "Gaussian Basis Sets for Use in Correlated Molecular Calculations. I. The Atoms Boron through Neon and Hydrogen". In: *The Journal of Chemical Physics* 90.2, pp. 1007–1023. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.456153. (Visited on 10/28/2026).

Kim, Yong-Ki and M. Eugene Rudd (Nov. 1994). "Binary-Encounter-Dipole Model for Electron-Impact Ionization". In: *Physical Review A* 50.5, pp. 3954–3967. ISSN: 1050-2947, 1094-1622. DOI: 10.1103/PhysRevA.50.3954. (Visited on 10/28/2026).

Hwang, W., Y.-K. Kim, and M. E. Rudd (Feb. 1996). "New Model for Electron-Impact Ionization Cross Sections of Molecules". In: *The Journal of Chemical Physics* 104.8, pp. 2956–2966. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.471116. (Visited on 10/28/2026).

Ali, M. A. et al. (June 1997). "Electron-Impact Total Ionization Cross Sections of Silicon and Germanium Hydrides". In: *The Journal of Chemical Physics* 106.23, pp. 9602–9608. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/1.473842. (Visited on 10/28/2026).

Huo, Winifred M and Yong-Ki Kim (Mar. 2000). "Use of Relativistic Effective Core Potentials in the Calculation of Total Electron-Impact Ionization Cross-Sections". In: *Chemical Physics Letters* 319.5-6, pp. 576–586. ISSN: 00092614. DOI: 10.1016/S0009-2614(00)00150-0. (Visited on 10/28/2026).

Scott, Gregory E. and Karl K. Irikura (Nov. 2005). "Electron-Impact Ionization Cross Sections of Molecules Containing Heavy Elements (Z > 10)". In: *Journal of Chemical Theory and Computation* 1.6, pp. 1153–1161. ISSN: 1549-9618. DOI: 10.1021/ct050077j.

Weigend, Florian and Reinhart Ahlrichs (2005). "Balanced Basis Sets of Split Valence, Triple Zeta Valence and Quadruple Zeta Valence Quality for H to Rn: Design and Assessment of Accuracy". In: *Physical Chemistry Chemical Physics* 7.18, p. 3297. ISSN: 1463-9076, 1463-9084. DOI: 10.1039/b508541a. (Visited on 10/28/2026).

Jiao, C Q, C A DeJoseph, and A Garscadden (Jan. 2007). "Electron Impact Ionization and Ion Reactions in N-Butane". In: *Journal of Physics D: Applied Physics* 40.2, p. 409. ISSN: 0022-3727. DOI: 10.1088/0022-3727/40/2/018. (Visited on 10/22/2026).

Ali, M A and Yong-Ki Kim (July 2008). "Ionization Cross Sections by Electron Impact on Halogen Atoms, Diatomic Halogen and Hydrogen Halide Molecules". In: *Journal of Physics B: Atomic, Molecular and Optical Physics* 41.14, p. 145202. ISSN: 0953-4075, 1361-6455. DOI: 10.1088/0953-4075/41/14/145202. (Visited on 10/28/2026).

Zheng, Jingjing, Xuefei Xu, and Donald G. Truhlar (Feb. 2011). "Minimally Augmented Karlsruhe Basis Sets". In: *Theoretical Chemistry Accounts* 128.3, pp. 295–305. ISSN: 1432-881X, 1432-2234. DOI: 10.1007/s00214-010-0846-z. (Visited on 10/28/2026).

Dolg, Michael and Xiaoyan Cao (Jan. 2012). "Relativistic Pseudopotentials: Their Development and Scope of Applications". In: *Chemical Reviews* 112.1, pp. 403–480. ISSN: 0009-2665. DOI: 10.1021/cr2001383. (Visited on 10/28/2026).

Kumar, Rajeev (Aug. 2014). "Electron Ionization Cross Sections for the PH3 Molecule". In: *Chemical Physics Letters* 609, pp. 108–112. ISSN: 00092614. DOI: 10.1016/j.cplett.2014.06.034. (Visited on 10/22/2026).

Hellweg, Arnim and Dmitrij Rappoport (Jan. 2015). "Development of New Auxiliary Basis Functions of the Karlsruhe Segmented Contracted Basis Sets Including Diffuse Basis Functions (Def2-SVPD, Def2-TZVPPD, and Def2-QVPPD) for RI-MP2 and RI-CC Calculations". In: *Physical Chemistry Chemical Physics* 17.2, pp. 1010–1017. ISSN: 1463-9076. DOI: 10.1039/c4cp04286g. (Visited on 10/28/2026).

Wang, Lee-Ping and Chenchen Song (June 2016). "Geometry Optimization Made Simple with Translation and Rotation Coordinates". In: *The Journal of Chemical Physics* 144.21, p. 214108. ISSN: 0021-9606. DOI: 10.1063/1.4952956. (Visited on 10/30/2026).

Sun, Qiming et al. (Mar. 2017). *The Python-Based Simulations of Chemistry Framework (PySCF)*. DOI: 10.48550/arXiv.1701.08223. arXiv: 1701.08223 [physics.chem-ph]. (Visited on 10/28/2026).

Bennett, M. Chandler et al. (Sept. 2018). "A New Generation of Effective Core Potentials from Correlated Calculations: 2nd Row Elements". In: *The Journal of Chemical Physics* 149.10, p. 104108. ISSN: 0021-9606. DOI: 10.1063/1.5038135. (Visited on 10/29/2026).

Wang, Guangming et al. (Oct. 2019). "A New Generation of Effective Core Potentials from Correlated Calculations: 4s and 4p Main Group Elements and First Row Additions". In: *The Journal of Chemical Physics* 151.14, p. 144110. ISSN: 0021-9606. DOI: 10.1063/1.5121006. (Visited on 10/29/2026).

Graves, Vincent, Bridgette Cooper, and Jonathan Tennyson (Mar. 2021). "The Efficient Calculation of Electron Impact Ionization Cross Sections with Effective Core Potentials". In: *The Journal of Chemical Physics* 154.11, p. 114104. ISSN: 0021-9606, 1089-7690. DOI: 10.1063/5.0039465. (Visited on 10/28/2026).

Gaches, Brandt A. L. et al. (Apr. 2024). "The Astrochemistry Low-Energy Electron Cross-Section (ALeCS) Database - I. Semi-Empirical Electron-Impact Ionization Cross-Section Calculations and Ionization Rates". In: *Astronomy & Astrophysics* 684, A41. ISSN: 0004-6361, 1432-0746. DOI: 10.1051/0004-6361/202348293. (Visited on 10/21/2026).

Burgasser, Adam J. et al. (Nov. 2025). "Observation of Undepleted Phosphine in the Atmosphere of a Low-Temperature Brown Dwarf". In: *Science* 390.6774, pp. 697–701. ISSN: 0036-8075, 1095-9203. DOI: 10.1126/science.adu0401. arXiv: 2510.03916 [astro-ph.SR]. (Visited on 10/28/2026).

Demuth, Tori, Julian Schnizer, and Dennis Svatunek (May 2026). "strainedSMILES2xyz: A Workflow for Reliable 3D Structures of Strained Molecules from SMILES". In: *Journal of Cheminformatics* 18.1, p. 80. ISSN: 1758-2946. DOI: 10.1186/s13321-026-01219-0. (Visited on 10/30/2026).